

Computer Architecture

Course No. 21472

Week #1

Introduction & Orientation

Bumsoo Kim

School of Computer Science & Engineering

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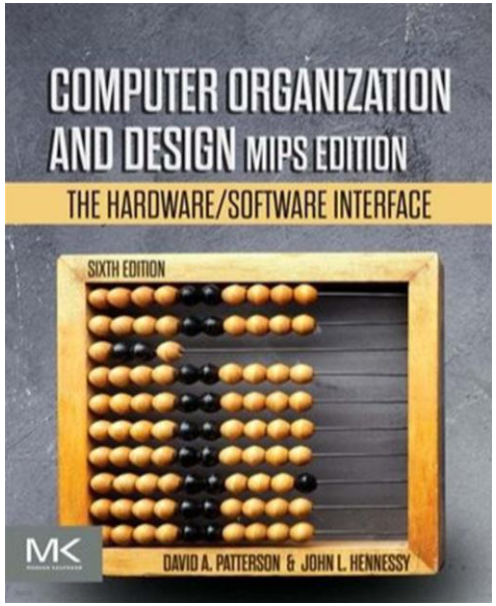
Office : 310-430

About this Course

Class Information

- Course Name : Computer Architecture
- Course No. : 21472
- Instructor: Bumsoo Kim
- Classroom : 310-728
- All class will be fully provided in English.
- Questions must be done via appointments. Will share informative questions with the rest of the class in documents.

Textbook



Computer Organization and Design (MIPS Edition) - The hardware/software interface
- David A. Patterson, John L. Hennessy

- One of the most standard textbooks for Computer Architecture courses in top schools such as Stanford, Harvard, SNU, and CAU.
- Reading the textbook will provide more sufficient understanding of this course.

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About



Bumsoo Kim (김범수) Ph.D.

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Assistant Professor

School of Computer Science & Engineering

Chung-Ang University (2024.09~)

Research Keywords

Real Time Human-Object Interaction

Language-Image Alignment (Hallucination)

Understanding 3D Scenes with Language

Medical Image Analysis

Work Experience

2020.08~2022.03 Kakao Brain (Video Scene Graph Search)

2022.03~2024.08 LG AI Research (Language-Image Pretraining, Medical Imaging)

Website

https://www.multilab.cau.ac.kr

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Computer Architecture, Why Bother?

Debugging

- Can diagnose when the application behavior is not running as expected.
- Demystify errors that can occur while attempting to write applications on hardware.

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Efficiency

- Enable faster / smaller / less prone to error programming.
- Allows us to assess relative costs of operations.
- Enables programmers to assess the the effect of programming choices.

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But most of all…

- It's usually a mandatory course in most CSE departments…

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Syllabus

Timeline

Week	Topic
#1	Introduction & Orientation
#2	Performance
#3	Instructions: Language of Computer
#4~#5	Arithmetic for Computers
#6~#7	Processor
#8 Mid-term Exam	
#9~#11	Pipelining
#12~#14	Memory Hierarchy
#15	Introduction to GPU
#16 Final Exam (Cumulative)	

This course may be quite intense before mid-term, but it will get easier. Don't be scared ☺

Will have occasional classes where I introduce my research focus (multimodal artificial intelligence covering computer vision and natural language processing). Feel free to doze off, eat snacks, or do your own work if you are not interested.

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Gradings

Assessment Items & Ratio

[4] 학습 평가 방법(Student Assessment)		
평가 항목(Assessment Item)	평가 비율(%)(Assessment Ratio)	추가 설명(Additional Description)
출결(Attendance)	10	More than 4 absence in class will result in an F grade.
중간시험(Mid-term Exam)	45	closed-book offline exam
기말시험(Final Exam)	45	closed-book offline exam

- Attendance is important! (and this class starts freakishly early, so think twice before you attend this course)
- Attendance will be randomly checked during class. It will NOT be pre-notified, and those who were absent during attendance check will NOT be reimbursed under ANY circumstances.
- If being absent for an unfortunate event that turns up, you will have to send me an e-mail BEFORE class. There will NEVER be accepted afterhand.

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About Class

Class will be consisted of

- The course meets for a total of 3 hours (180 minutes) per week, including two 45-minute lectures, two 15-minute breaks, and a 60-minute office hour (10:00 am~11:00 am).
- The office hour will be held every week at my office (310관 430호).
- All class materials will be provided only in English, but during office hours you are free to use Korean.
- You exams will be taken in ENGLISH!!
- ALL of your questions should be addressed during the office hour. I will **NOT** respond to e-mails unless they are critical or urgent issues regarding this class.

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Thank you

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